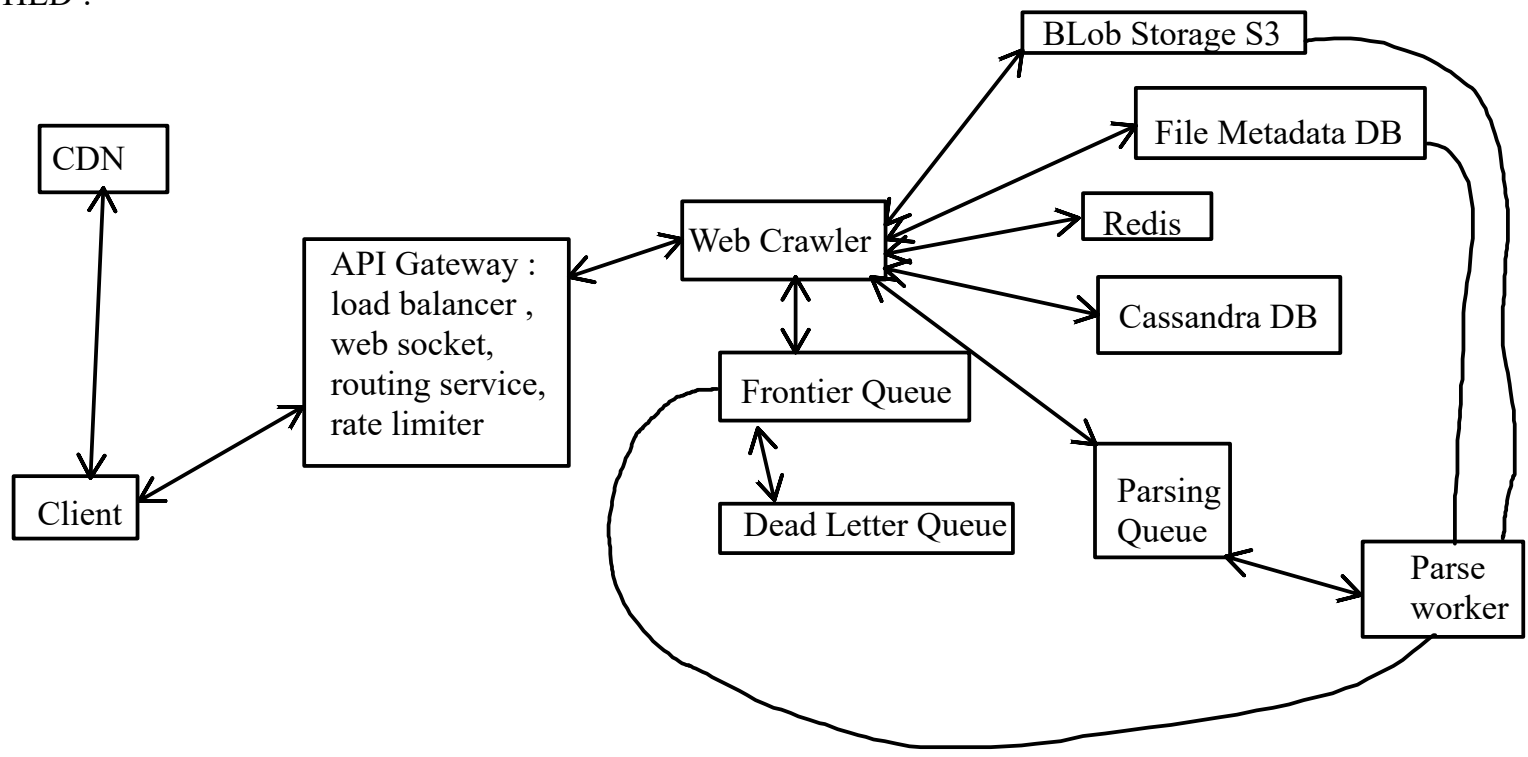
Web Crawler

Functional requirements:
user should be able to extract/download from original metadata ,
user should be able to place multiple downloads ,

Non-functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (Here eventual consistency is fine , but here we need 99.99% uptime
i.e Availability>>Consistency)

Core Entities:
user,
filemetadata,
history,
Activity

API :
POST:/extract/download
{
    userID,
    {filemetadata}
}

HLD :